

container-death-triage

Container Death Triage — telling six lookalikes apart in one step

Revision: 1 **Last modified:** 2026-08-21T17:06:26Z **Audience:** Anyone who has just found a boba container stopped, restarted, or missing, and needs to know *what killed it* before proposing a fix. **Authority:** Derived from BOB-131, where one ticket recorded “podman common crash” for two unrelated events, neither of which was a common crash. Full evidence: <docs/incidents/2026-08-21-bob131-container-death-triage.md>. **Tool:** scripts/diagnostics/bob131_container_death_triage.sh

Start here

```
$ ./scripts/diagnostics/bob131_container_death_triage.sh
$S collect qbittorrent-proxy "2026-08-20 17:55:30" "2026-08-20
    17:57:00" > /tmp/bundle.txt
$S classify /tmp/bundle.txt
```

Timestamps are **host-local** (journalctl syntax). The tool is read-only: it never starts, stops, restarts, removes or signals anything.

The decision table

Read **top to bottom** and stop at the first row that matches. The order is not cosmetic — the classes overlap in symptoms, and the cheap signals lie.

#	Class	The one signal that settles it	What it means
1	HOST-POWER-TRANSITION	systemd-logind: System is powering down / a boot boundary in journalctl --list-boots	The container did not fail. The host went down. Exit code is usually 0 , and <i>every</i> container dies at once.
2	THREAD-LIMIT-EXHAUSTION	EAGAIN, failed to create new OS thread, pthread_create failed: Resource temporarily unavailable, Cannot allocate memory on fork/clone	§12.12. Not an OOM. Compare <code>ulimit -u</code> with <code>ps -L --no-headers -u "\$USER" \ wc -l</code> .
3	CGROUP-OOM-KILL	memory.events: oom_kill > 0 , or kernel Memory cgroup out of memory naming libpod-<cid>	§12.6 containment working as designed. The container hit <i>its own</i> limit. Decode <code>oom_memcg:</code> <code>libpod-... = the container's cap;</code> <code>user@1000.service = session OOM.</code>
4	PROCESS-SIGSEGV	died exit=139 (128+11), and a kernel segfault / ANOM_ABEND ... sig=11	A real crash inside the container. Runtime is fine.
5	ORCHESTRATED-STOP	died exit=137 (128+9) ~10 s after a restart / stop / down event	SIGTERM-timeout SIGKILL. Not a crash, not an OOM. PID 1 was

#	Class	The one signal that settles it	What it means
			too slow to shut down.
6	CGROUP-MEMORY-CEILING	memory.events: max > 0 with oom_kill == 0	Pressure, never a cause of death. The kernel <i>reclaimed</i> ; it killed nothing.

The three that get confused, and why

oom_kill > 0 vs max > 0 — a kill vs a squeeze

Both live in the same file and read as “memory trouble”. They are not the same event.

```
CID=$(podman inspect qbittorrent-proxy --format '{{.Id}}')
CG=$(find /sys/fs/cgroup -maxdepth 8 -type d -name "libpod-${CID}.scope" | head -1)
cat "$CG/memory.events"
```

- **oom_kill > 0** → the kernel **killed** something at the cgroup limit.
- **max > 0, oom_kill 0** → the cgroup **reached** its limit and the kernel reclaimed/swapped instead. Nothing died of it.

Measured on this stack (mem_limit: 768m for download-proxy):

```
memory.max      805306368    # 768 MiB
memory.peak     805371904    # pinned at the ceiling
memory.events:  max 1584     # 1584 times in ~48 min
                  oom_kill 0    # ...and killed nothing
```

That container lives permanently against its ceiling and reports **healthy**. Citing max 1584 as a cause of death is wrong. It is worth its own ticket as *capacity*; it is not an explanation for a crash.

OOM-kill vs thread exhaustion — both say “cannot allocate”, neither is the other

Thread/RLIMIT_NPROC exhaustion (§12.12) fails clone(2) with **EAGAIN** while **memory is abundant**. Every memory metric looks healthy while it happens, so it is routinely misfiled as an OOM.

```
ulimit -u; ulimit -Hu          # the ceiling
ps -L --no-headers -u "$USER" | wc -l  # live threads for
    this UID
```

Current headroom on this host: limit **65536**, observed peak **1559** (2.4 %). Thread exhaustion is **not** currently plausible here — it was, historically, when the limit was 4096.

Never use `kill -f / killall / pgrep -f` to investigate: the pattern matches your own command line (§11.4.263, §12.12). And never `signal pgid ≤ 1`.

“common crashed” — almost always a carrier match

`common <error>`: ... in the log means common **reported** a fault. A common *crash* would put common itself in a kernel `segfault / ANOM_ABEND` line.

```
# the thing, not a mention of it:
journalctl --since "<window>" | grep -E
    'kernel: common\[ [0-9]+\]: segfault|
    ANOM_ABEND.*comm="common"'
```

Empty ⇒ common did not crash, whatever the `<error>` lines say. These two are benign races during teardown, not faults:

```
common <CID> <error>: Failed to write 137 to exit file: ... No
such file or directory
common <CID> <error>: Failed to create container: exit status 1
```

Likewise `common <nwarn>`: Failed to open cgroups file: `/sys/fs/cgroup/memory.events` is ordinary rootless noise — **50 290** occurrences in one week here.

Gotchas that will waste your time

- **Timezones disagree.** `podman ps` / some podman events records print **+0300** for boba containers while the host journal prints **+0200 CEST**. Always convert to UTC before correlating: `date -u -d "@<epoch>".`
- **podman events --since/--until may return nothing** for a window that plainly contains events. The journal is authoritative; the tool falls back to it and does not depend on the events stream for its verdict.
- **cgroup files are LIVE.** For a historical window they describe *today's* container carrying that name, not the dead one. The bundle labels this.

- **A removed container's podman logs are gone**, but its stdout survives in the journal because the log driver is journald:
`journalctl CONTAINER_NAME=qbittorrent-proxy --since ... --until ...` This is how BOB-131's crash was read after the container was long gone.
 - **journalctl | grep over the full retention takes minutes**. Always bound it with `--since`, and prefix long runs with `nice -n 19 ionice -c 3`.
 - **find here is bfs, not GNU findutils**. Relative `-newermt` forms are rejected — they error to stderr and print nothing to stdout.
 - **grep -q and awk '...; exit' inside a pipeline** close the pipe early and SIGPIPE the writer; under `set -euo pipefail` that aborts your script, and it does so *racily* (§11.4.201(12)). Drain instead:
`grep -E ... >/dev/null, awk '...' | head -1`.
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Boot persistence — why a container can be missing with nothing wrong

restart: unless-stopped is a within-podman policy. **It does not survive a host power cycle**. On this host, as measured 2026-08-21:

```
podman-restart.service    disabled    (user and system scope)
boba-stack.service        linked, disabled
Linger=yes
```

So after any power-off the boba stack stays down until someone runs `./start.sh`. In BOB-131 that was **14h 06m 43s**. If a container is *absent* rather than *unhealthy*, check the boot boundary before looking for a crash.

Verifying the tool still has teeth

```
./scripts/diagnostics/bob131_container_death_triage.sh --selftest
```

Eight golden fixtures — one per class, a negative control that must stay UNKNOWN, and a common-carrier fixture that must **not** change the verdict. If any mutation of the classifier leaves this green, the check is decorative and must be repaired before it is trusted (§1.1, §11.4.115(F)).